

UNIT-4

STRUCTURED QUERY LANGUAGE

1. Write about SQL and its History?

SQL: It is a structured query language, which is a computer language for storing, manipulating, and retrieving data stored in a relational data base.

It includes database creation, fetching rows, modifying rows etc..

- It is a programming language that is used to work with data base.
- It is an ANSI (AMERICAN NATURAL STANDARD INSTITUTE) standard language.
- It has a different version which is used for relational data base systems.
- All the related data base management system (RDBMS) like MY SQL, MS-ACCESS, Oracle, Sybase, Postgre sql and SQL Server use SQL as their standard data base language.

Advantages

SQL is widely popular because it offers following advantages:-

- It allows user to access data in RDBMS
- It allows user to describe the data
- It allows user to define the data in a database and manipulate the data.
- It allows user to create and drop database and tables.
- It allows user to set permissions on tables, procedures and views.
- It allows to embed within other languages using SQL modules, libraries etc..
- It allows to user to create tables, views, sequences, indexes etc..

HISTORY OF SQL:-

SQL is initially developed in IBM by DONOLD AD. CHAMBERLIN and RAYMOND F BOYLE in 1970.

1.1970:-

- DR.E.F.CODD published the paper “ A relational model of data for large shared data banks” .
- CODD’S model is now accepted as a definitive model for relational data base management system(RDBMS).

2.1978:-

- **IBM worked** to developed codd’s ideas and released a product name **SYSTEM/R**.
- Later the language structured English query language (SEQUEL) was developed by **IBM** corporation to use cod’s models.

3.1979:-

- **SEQUEL** later it became SQL in 1979.
- Today, sql is accepted as the standard RDBMS language

4.1986:-

The first relational database was first released by relational software which later came to be known as “ORACLE”.

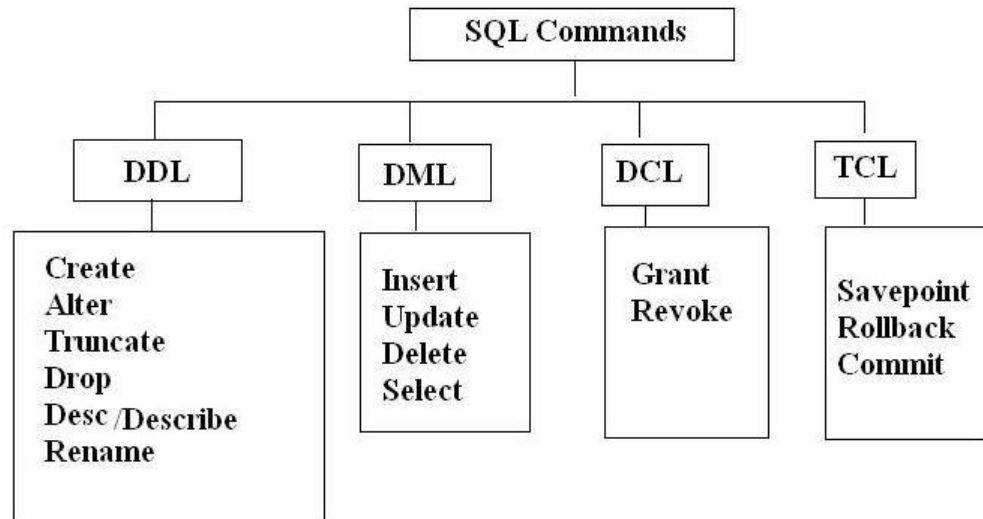
2. Explain about different types of SQL commands?**SQL Commands:**

SQL is a software which is used to work with database. It is a fourth generation languages.

It is a non-procedural language which contains set of commands.

All SQL commands are divided into four types. They are

1. Data definition language (DDL).
2. Data manipulation language (DML).
3. Data control language (DCL).
- 4.Transaction Control language(TCL)



1. DDL:

DDL commands are used to define the different structures of data bases such as tables, views, sequences, etc...

The common DDL commands are

Command	Description
Create	Create a new table, a view of a table or other objects in a database.
Alter	Modifies an existing data base object, such as a table.
Drop	Deletes an entire table or view or other objects in the data base.
Rename	Change the name of the existing table.
Truncate	Delete the entire data in a table.
Comment	Put comments on existing table.

2. DML:

DML commands are used to modify or insert or retrieve data in a table.

COMMANDS	DESCRIPTION
Select	Retrieve the certain records from one or more tables.
Insert	Create a record.
Update	Identify the records.
Delete	Deletes the record.

3.DCL

This commands used to give privileges to access limited data

Command	Description
Grant	Gives a privileges to user.
Revoke	Takes back privileges back from users .

4.TCL

TCL are used to apply the changes permanently to the saved data in the data base.

Command	Description
Commit	Perment work saved into data base.
Roll back	Restore data base to original form,since the last commit.
Save point	Create a save point for later use rollback the new change.
Set transaction	It set that transaction properties such as react write/ read only access.

3.Explain about datatypes in SQL?

A:Data types in SQL:

- In SQL,data types specifies what type of data a column can contain.
- SQL data type is an attribute that specifies type of data of any object.
- Data types are defines for columns of a table, local/global variances.

SQL has different types of datatypes. They are

1. Exact numeric data types
2. Approximate numeric data types .
3. Date and time data type.
4. Character string data types
5. Other data types.

1.EXACT NUMERIC DATATYPES:-

There are different types of exact numeric data types which are used to accept integer values.

DATA TYPES	FROM	TO
Int	-2,147,483,648	+2,147,483,647
Small int	-32,768	+32,767
Big int	922,337,203,685,477,5808	922,337,203,685,477,5807
Tiny int	0	255
Bit	0	1
Decimal	-10 ³⁸ +1	+10 ³⁸ +1
Numeric	-10 ³⁸ +1	10 ³⁸ +1
Money	922,337,203,685,477,5808	+922,337,205,685,477,5807
Small money	-214,748,3648	214,748,3647

INT:-

1. The integer datatype accepts numeric value.
2. It stores any type of integer values either positive(or)negative.
3. The range is -2,147,483,648 to +2,147,483,647

The different types of integer data types are:

- Big int
- Small int
- Tiny int
- Bit

Ex:- s.no int ,s.no integer , s.no bigint.

DECIMALS :-

It accepts numeric values, which may define **precision & scale** in the data type declaration.

SYNTAX:- decimal(p,s)

- P-precision it represents the total no.of digits.
- S-scale it represents the number of digits accepted after decimal point.

EX:- avg decimal(5,3)

Numeric:- It is similar to decimal which accepts both precision and scale.

SYNTAX:- numeric (p,s)

MONEY:- This datatype is used to store money/currency information.

2. APPROXIMATE NUMERIC DATA TYPES:-

These data type is used to accept floating point values.

Data types	from	to
Float	-1.79E+308	1.79E+308
Real/double	-3.40E+38	3.40E+38

FLOAT :-

- It accepts real values which may define a precision of upto a maximum of 64.

SYNTAX :-

Float(p)

Ex :- avg float(10)

REAL / DOUBLE :-

- it accepts real values of upto a precision of 64.
- no parameters are required when declaring a real type.

SYNTAX :- Real Ex :- avg real.

3.DATE AND TIME DATA TYPES :-

These are used to accept date and time formats.

Data type	from	To
Date time	Jan 1,1753	Dec 31,9999
Date	Stores a date like jun 31,1991	
Time	Stores time of day like 12:00pm	

4.CHARACTER STRING DATA TYPES :-

It accepts character type data. The different types of character data types are

1.char

2.varchar-variable character

3.varchar2

1.char:-

- it accepts a fixed length non – Uni code character data .
- it stores a maximum length upto 8000 characters.

Syntax :- char(n)

Here 'n' represents number of characters .

Ex :- s.name char(10)

2.Varchar :-

- it accepts a variable length , non –unicode character data.
- It stores a maximum length upto 4000 characters .

Syntax :- varchar(n)

Ex :- s.name varchar(10).

3.varchar2 :-

- It accepts a variable length non -Unicode character data.
- It stores a maximum length up to 8000 characters .

Syntax :- varchar2(n)

Ex :- s.name varchar(10).

Other data types :-

The other data types are CLOB (CHARACTER LARGE OBJECT) ,BLOB (BINARY LARGE OBJECT)

CLOB :-

IT stores variable length character data that can be a single byte (or) multi – byte
(supports more than 4 gb)

BLOB :-

This type is used to store large amount of binary data such as images (or) other type of files .

4Q) EXPLAIN ABOUT DDL COMMANDS ?**A) SQL :-**

- **SQL** stands for **STRUCTURED QUERY LANGUAGE**.
- It is a software which is used to work with databases.
- **It** is a fourth generation language.
- it is used for storing manipulating and retrieving data stored in a relational data base.

DDL COMMANDS :-

DDL stands for DATA DEFINITION LANGUAGE . DDL commands are used to work with the structure of the data base .

There are six different commands in DDL. They are

1.create

2.Alter

- 3.drop
- 4.rename
- 5.truncate
- 6.comment

1.create :-

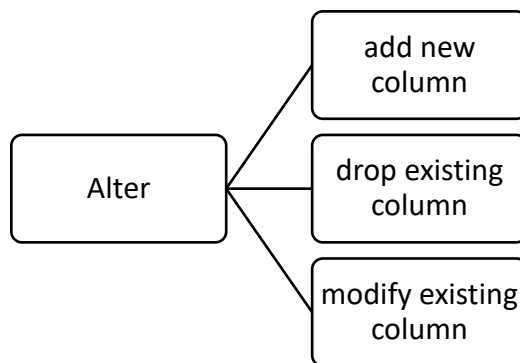
- Create command is used to create data base object like tables , indexes, views, sequences and soon on data base .

Syntax :- Create table ,tablename (column datatype(size),column2 datatype(size).....);

Ex :- command for creating customer table .

SQL>create table customer (eno varchar(5),ename varchar(20), Addres varchar(20));

2.ALTER :- Alter command is used to modify the structure of data base objects like tables .

**ALTER COMMAND TO ADD NEW COLUMN :-**

SYNTAX :- Alter table tablename add (columnname datatype(size));

Ex :- Alter table customer ADD (phnno char(10),email id varchar(20));

ALTER COMMAND TO DROP A COLUMN:-

SYNTAX :- Alter table tablename drop (course name);

Ex :- Alter table customer Drop (emailid);

ALTER COMMAND TO MODIFY:-

SYNTAX :- Alter table tablename modify(column datatype(size),.....);

Ex :- SQL>Alter table book modify(ISBN varchar(12));

3.DROP :-

- Drop command is used to destroy (delete) an existing object from a database.

Syntax :- Drop table tablename ;

EX :- SQL>Drop table book ;

4.RENAME :-

- Rename command is used to rename a table .

Syntax :- Rename<old tablename>to<new tablename>;

Ex:- SQL>Rename employee to emp;

5.TRUNCATE:-

- This command is used to delete complete data in a table . The data deleted from tables cannot be rolled back.

Syntax :- truncate table <table name>;

Ex:- SQL>truncate table emp;

6.COMMENT :-

- This command is used to add comment to the existing table.

Syntax :- Comment on table <tablename>is “comment sentence”.

Ex :-comment on the student is “to store student personal details”;

To view the table comment :-

Syntax :- SQL >select comments from user _tab_comments where
Table-name ='TABLE NAME';

*the table name must be given in capital letters only.

5) EXPLAIN ABOUT DML COMMANDS ?

A) **SQL :-** refer sql introduction in previous question .

DATA MANIPULATION LANGUAGE :-

DML is used to insert, modify, delete and retrieve data in a data base.

There are four types of DML commands are available in SQL. They are

- 1.INSERT
- 2.UPDATE
- 3.DELETE
- 4.SELECT

1.INSERT:- This command is used to add records to the existing table.

- “&” is used to insert command to accept values at SQL prompt.
- There are three ways to insert records in a table.

To insert values for one record :-

Syntax:- insert into tablename values(value1,value2.....);

To insert data for a particular columns in a table :-

Syntax :- insert into tablename ((columnname1,columnname2....) values(exp1,exp2....);

To insert multiple records at a time :-

Syntax:- insert into tablename values (&columnname1.....);

NOTE:-Put single quotes (' ') for varchar data type column in insert command.

Ex :- SQL>insert into vijaya values(&sno,'&sname',&marks);

Enter value for s.no: 1

Enter value for sname : lucky

Enter value for marks : 98

1 row created

SQL>/ Here '/' is a character used to enter another record in a table.

2.UPDATE :-

i) The update command is used to change (or) modify data values in a table .

ii) it is used to update either all rows from a table.(or)set of selected rows from a table.

Syntax :- update <tablename> set colname = <expression>, colname=<expression>;

- This command is used to update all rows in a table

Syntax :- Update <tablename> set colname=<expression>,colname = <expression>

[where <condition>;

- To update selected row in a table

Ex :- Update student set tot =sub1+sub2+sub3;

(or)

Update student set tot = sub1+sub2+sub3 where s no=2;

3.DELETE :-

- The delete command is used to delete the records from a table.
- It is used to delete either all rows from a table (or)set of selected rows from a table.

Syntax :- to delete all rows from a table

Delete from <tablename>;

- **To delete a particular row in a table .**

Syntax :- delete from<tablename>[where < condition>;

ex :- SQL>delete from student;

(or)

SQL> delete from student where sno =2;

4.Select :- Select command is used to retrieve the data from a table in data base according to user requirements .

The complete syntax for select statement is :

Syntax :-

Select[distinct][*][colnames] [aggregate operation] from <tablename>.

[where <search condition>]

[Group by group –by-us]

[Having<search condition>]

[order by orderlist [asc][desc]];

Ex :- select *from student;

(or)

Select sub1,sub2,sub3 from student ;

(or)

Select *from student where sno=3;

6) Difference Between Truncate And Delete ?

TRUNCATE	DELETE
1. Truncate is a DDL command.	1. Delete is a DML command .
2. it is used to delete complete data in a table leaving the table empty.	2. it is used to delete record from a table .
3. syntax :- Truncate table <TN>;	3.syntax :- Delete from<TN> [where <condition>;]
4.ex:- SQL>Truncate table sales;	4.ex :- SQL>Delete from customer Where eno ='co3';
5. Rolled back cannot be applied to Truncate.	5. Rolled back can be applied to delete.

7) WRITE ABOUT AGGREGATE FUNCTIONS OF SQL?

A)SQL:- refer SQL introduction in previous question.

SQL FUNCTIONS :-

SQL functions are used to perform different types of operations on data in a data base .

There are five different types of SQL functions . They are

- NUMERIC FUNCTIONS
- STRING FUNCTIONS
- AGGREGATE FUNCTIONS
- DATE FUNCTIONS

1.AGGREGATE FUNCTIONS :-

- These are the functions in SQL which are applied in a group of records.
- These are used to find aggregate values like sum, average etc...for a group of records in a database
- We can apply this function on a database by using a special keyword known as '**GROUP BY**'.
- **GROUP BY** is a keyword which is used to group the records in a database on a particular column.
- The different types of aggregate functions are
 - 1.COUNT.
 - 2.SUM.
 - 3.AVG.
 - 4.MAX.
 - 5.MIN.

1.COUNT:- The function is used to count a number of records in a database (table)

Ex:- SQL>select dno,count(*)from emp group by dno;

OUTPUT :-

DNO	COUNT(*)
10	5
20	4
30	2

2.SUM:- This function is used to get the sum of values for the given attribute on the database .

Ex:- SQL>select dno,sum(sal)from emp group by dno;

OUTPUT :-

DNO	SUM(SAL)
10	14000
20	90000
30	65000

3.AVG :- This function is used to get the average of values of given attribute from a database.

Ex :- SQL>select dno,avg(sal) from emp group by
dno;

OUTPUT

DNO	AVG(SAL)	:-
10	30000	
20	22500	
30	31000	

4.MAX :- This function is used to get maximum value for the given attribute form a database.

ex :- SQL>select dno,max(sal) from emp group by dno;

OUTPUT :-

DNO	MIN(SAL)
10	15000
20	35000
30	22000

5.MIN :- The function is used to get minimum value for a given attribute from a datatype.

Ex :- SQL>select dno,min(sal) from emp group by dno;

OUTPUT :-

DNO	MIN(SAL)
10	5000
20	20000
30	40000

8. Write about select operation in SQL?

A. Select: Select command is used to retrieve the data from a table in database according to user requirements.

The computer syntax for select statement is

Syntax:

Select[distinct][*][colnames][Aggregate operations] from <tablename>;

[WHERE <search condition>]

[GROUP BY group_by_list]

[HAVING <search condition>]

[ORDER BY orderlist[asc][desc]];

Distinct clause:

Distinct clause the SQL Distinct clause used along with the SELECT command. It is a keyword which retrieves only unique data entries depending on the column list.

Ex: select distinct student from emppayroll;

Or

Select distinct from tablename;

(*): Star:

It retrieves all the record in a table.

Ex: select * from emp;

WHERE:

- Where clause – The Where clause is used to filter records.
- The WHERE clause is used to extract only those records that fulfills a specified condition.

Ex: select * from emp where eno=2;

Group by clause:

- It is used to retrieve the records depending upon a particular column.
- It divides the column data into different groups.
- It is often used with aggregate functions to group the results with by one or more column.

Ex: select dno, sum(sal) from emp group by dno;

Having clause:

- The HAVING clause was added to SQL because the WHERE keyword could not be used with aggregate functions.
- It is used with group by statement.

Ex: SQL>select dno, sum(sal) from emp group by dno having dno>10;

Order by: The order by keyword is used to sort the data in ascending or descending order.

- It sorts the records in ascending order by default. To sort the record in descending order, we use 'Desc' keyword.

Ex: SQL> select * from emp order by eno;

Or

SQL> select * from emp order by eno desc;

9. Explain about various constraints in SQL**A. SQL constraints:**

- SQL constraints are used to specify rules for data in a table.
- These are used to limit the type of data that can go into a table to maintain the accuracy and integrity of the data inside table.

Constraints can be divided into following two types:

1. Column level constraints
2. Table level constraints

Column level constraints:

The constraints which are defined at column level are called as column level constraints.

Syntax: Create table <tablename> (colname1 datatype constraint name);

Table level constraints:

The constraints which are defined at table level are called as table level constraints.

Syntax: Create table <tablename> (colname1 datatype, colname2 datatype, constraint name(colname));

- Constraints are used to make sure that the integrity of data is maintained in the database
- The most used constraints that can be applied in a table are:

1. NOT-NULL constraint
2. UNIQUE constraint
3. PRIMARY KEY constraint
4. FOREIGN KEY constraint
5. CHECK constraint
6. DEFAULT constraint

NOT-NULL constraint:

- This constraint restricts a column from having a null value.
- Once not null constraint is applied to a column, we cannot pass a null value to that column.
- It can be applied at column level only.

Syntax: column-name datatype (size) NOT NULL;

Ex: SQL> create table stud_pers(stno number(2), sname varchar2(20) NOTNULL);

UNIQUE constraints:

- It ensures that a column will only have unique values.
- A unique constraint field(column) willnot have duplicate data.
- It can be applied at column level or table level.

Syntax: (At column level) Columnname datatype(size) UNIQUE;

Ex: SQL> create table stud_pers (stnonumber(2), sname varchar2(20) unique, add varchar2(30));

(At table level) UNIQUE (colname, colname.....);

EX: SQL> create stud_pers(sno number(2), sname varchar2(20) unique(sno));

PRIMARY KEY:

- It uniquely identifies each record in a database.
- A primary key must contain unique value.
- It is used to index the data inside the table.
- A table can have only one primary key which may consists of single or multiple fields.

Syntax: (At column level) Colname datatype(size) PRIMARY KEY;

Ex: SQL> create table stud_pers (snonumber(3) primary key, sname varchar(10));

- When multiple columns are defined as primary key, then it is called as composite key.
- These are defined at table level.

Syntax: (At table level) PRIMARY KEY (colname1, colname2.....);

Ex: SQL> create table stud_pers (snumber(3), sname varchar(10), primary key(sno, sname));

FOREIGN KEY:

- A foreign key is used to establish a link between two tables.
- A foreign key in one table used to point primary key in another table.
- It is also called as reference key.
- It comes under referential integrity.
- It can be applied at column level and table level.

Syntax: (At column level) Columnname datatype(size) REFERENCES old tablename(columnname)

Ex: SQL> create table stud_persmarks(stno number(2) references stud_pers(stno), sub1 number(3), sub2 number(3), sub3 number(3), total number(5));

(At table level)

Foreign key(colname) references tablename(colname);

Ex: SQL> create table stud_marks(stno number(3), sub1 number(3), sub2 number(3), total number(5), avg number(7,2) foreign key(stno references stud_pers(stno));

CHECK:

- It is used to restrict the value of a column between range.
- It evaluates the logical expression either true or false.
- It can be applied at both column level and table level.

Syntax: (At column level)

Columnname datatype(size) check (logical expression)

Ex: SQL> create table stud(sno number(2), sname varchar(10) check(sname like 's%'));

(At table level)

CHECK (logical expression)

Ex: SQL> create table stu(sno number(3), sname varchar(10) check(sno between 1 and 10));

DEFAULT:

The default constraint provides default values when no value is specified for a column.

Ex: SQL> create table emp(eno number(3), ename varchar(20), sal number(10) default 5000);

10. Explain about different types of joins in SQL.

A. Join:

SQL join is used to fetch the data from one or more tables.

- A join clause is used to combine rows from two or more tables based on a related column between them.
- Join is a keyword used in SQL queries for joining two or more tables.
- Join returns common values which are common to both the tables.

Ex: Demo table

Table1 – order

Orderid	customerid	orderdate
1	1	12-Aug-87
2	2	8-Aug-99
3	3	28-Sep-97

Table2 – customers

Customerid	customername	country
1	Madhu	U.S
2	Sri	Mexico
3	Vijju	Japan

Here, customerid is common to both the tables

Types of joins:

The different types of joins are:

1. Inner join
2. Left / left outer join }
3. Right / right outer join equi join
4. Full outer join
5. Self join

1.Inner join:

- Inner join keyword selects records that have matching values in both the tables.

Syntax: select column_name(s) from table1 inner join table2 on table1.column_name = table2.column_name;

From the above table,

Ex: SQL> select orders.orderid, customers.customername from orders inner join customers on order.customerid = customers.customerid;

RESULT: orderid	customername
1	Madhu
2	Sri
3	Vijju

2.Left join:

- The left join keyword returns all records from the left table(table1) and the matched records from the right table(table2).
- In some databases, left join is called as left outer join.

Syntax:

Select column_name(s) from table1 left join table2 on table1.column_name = table2.column_name;

From the above Demo table,

Ex: SQL> select orders.orderid, customers.customername from orders left join customers on orders.customerid = customers.customerid;

RESULT: orderid	customername
1	Madhu
2	Sri
3	Vijju

3.Right join:

- The right join keyword returns all records from the right table(table2) and the matched records from the left table(table1).
- The result is null from the left side, when there is no match.
- In some databases, right join is called as right outer join.

Syntax:

Select column_name(s) from table1 right join table2 on table1.column_name = table2.column_name;

From the above demo table,

Ex: SQL> select orders.orderid, customers.customername from orders right join customers
on orders.customerid = customers.customerid;

RESULT:	orderid	customername
	1	Madhu
	2	Sri
	3	Vijju

4.Full Outer Join:

- The full outer join keyword return all records from left table and all records from right table.
- It also returns all records where there is a match in either left table or right table.

Syntax:

Select column_name(s) from table1 full outer join table2 on table1.column_name =
table2.column_name;

From the above demo table,

Ex: SQL> select orders.orderid, customers.customername from orders full outer join
customers on orders.customerid = customers.customerid;

RESULT:	orderid	customername
	1	Madhu
	2	Sri
	3	Vijju

5.self join:

If a table is joined to itself to retrieve information, then we call it as self join.

This is also known as inner join.

Ex: SQL> select orders.orderid, customers.customername from orders, customers where
orders.customerid = customers.customerid;

11.Write about Cross Join?

Q. Cross Join:

- It is also called as cartesian join.
- It returns the cartesian product of the set of records from two or more joined tables.
- It produces a result set, which is the number of rows in the first table multiplied by no of rows in the second table.
- This kind of result is called cartesian product.

- If where clause is used along with cross join, then it is similar to inner join.

Syntax:

Select table1.column1, table2.column2,.....from table1, table2,.....;

Ex: Consider emp and dept tables

SQL> select * from emp;

<u>Eno</u>	<u>ename</u>	<u>address</u>	<u>sal</u>	<u>dno</u>
2	Devi	KKD	10000	20
4	Devi	HYD	10000	10

SQL> select * from dept;

Dnodname

1	computers
20	maths

Cross join query:

SQL> select eno, enmae, sal, emp.dno, dept.dno, dname from emp, dept;

RESULT:

<u>Eno</u>	<u>ename</u>	<u>sal</u>	<u>dno</u>	<u>dno</u>	<u>dname</u>
2	Devi	10000	20	1	computers
4	Devi	10000	10	1	computers
2	Devi	10000	20	20	maths
4	Devi	10000	10	20	maths

12. Explain About Views In Sql ?

A. Views :-

1. In SQL , a view is a virtual table based on the result-set of an SQL statement.
2. A view contains a rows and columns just like a real table.
3. The fields in a view are fields from one or more real tables in a database.
4. It can be createdfrom one or many tables which depends on written sql query to create a view.

Creating views :-

Database views are created using the create view statement .

- Views can be created from
 1. Single tables
 2. Multiple tables
 3. Another view

1. Creating view from single table :-

The basic form of creating a view using a single table is as follows :-

Syntax :- Create view viewname as select column1,col2,..... From tablename ;

Ex :- SQL> create view v as select eno ,ename from emp ;

- To retrieve the data from 'view'

Syntax :- select * from viewname ;

ex :- select * from v ;

result :-

eno	ename
1	Prabhas
2	Arjun
3	Surya

2. Creating view from multiple tables :-

The basic form of creating a view using multiple tables is as follows :-

Syntax :- create view viewname as select table1.col1 , table2.col2, from table1,table2 ;

ex :- create view v1 as select emp.eno ,dept.dno from emp , dept ;

*To retrieve the data from view v1 .

Ex :- SQL>select * from v1 ;

Result :-

eno	dno
1	1
2	1
1	2
2	3

3.Creating a view from another view :-

The basic form of creating a view using another view is as follows :

Syntax :- Create view viewname as select col1,col2,..... from viewname ;

Ex :- SQL>select view v2 as select eno , ename from v ;

Updating views :-

- We can also update or add a column in an existing view .

Syntax :- Create or replace view viewname as select col1 , col2 from tablename ;

Ex :- SQL> create or replace view v as select eno , ename , sal from emp ;

(Or)

SQL> create or replace view v as select eno,ename,sal from emp where eno =2 ;

- To retrieve the data from view .

Ex:- select * from v ;

Result :- eno enamesal

1	Ale	25000
2	Sow	23000
3	Taraka	22500
4	Madhu	30000
5	Vijji	42000

Inserting data into view :-

- We can insert the data into view by using insert command like a real table .
- If the data is inserted in a view then it can be updated in real table also.

Syntax :-

Insert into viewname values (&col1 ,&col2 ,);

Deleting data from view :-

- We can delete entire data or particular records in view by using delete command like a real table .

Syntax :- Delete from viewname where [condition] ;

Ex :- * delete from v ;

(or)

*delete from v where eno =2;

Ex :- select * from v ;

Result :- eno enamesal

1	Akun	4200
2	Gouri	4800
3	Sri	3000

Dropping a view :-

The entire view structure can be deleted by using drop command .

Syntax :-

Drop view viewname ;

Ex :- SQL> drop view v ;

13. Explain how to add constraints using Alter ?

Ans :- **Constraints using alter command :-**

Alter command in sql is used to add (or) modify new (or) existing column and datatype .

Alter is also used to add (or) drop in a table after creation.

1. **Primary key :-** It uniquely identifies each record in a database .

Syntax :- Adding primary key using alter :-

Alter table tablename add constraints constraint_name primary key [column name] ;

(or)

Alter table tablename add primary key (column_name);

EX :- Alter table emp add constraint pk primary key (eno) ;

Here , pk is the constraint name .

Syntax :- Dropping primary key using alter :-

Alter table table_name drop constraint constraint_name ;

EX:- Alter table emp drop constraint pk ;

2.Foreign key:-

It is used to establish a line between two tables .

Syntax :-**Adding foreign key using Alter :-**

Alter table tablename add constraint constraint_name foreign key (col.name) references
name (name);

EX:-Alter table emp add constraints pk foreign key (dno) references emp (dno) ;

Syntax :-**For dropping foreign key :-**

Alter table tablename drop constraints constraint_name ;

EX :- Alter table emp drop constraints pk ;

3.Not Null :-

It restricts a column from having a null value .

Syntax :-**Adding NotNull using alter :-**

Alter table tablename add constraints constraints_name NotNull (col_name) ;

EX :- Alter table emp add constraint c NotNull (ename);

4.unique :-It enables that a column will only have unique values .

Syntax :- Adding unique using alter :-

Alter table tablename add constraints constraint_name unique (col_name);

EX :- Alter table emp add constraint c1 unique (eno) ;

Syntax :-**Dropping unique using alter :-**

Alter table tablename drop constraints constraints_name ;

EX :- Alter table tablename drop constraints c1 ;

5.Check :-It is used to restrict the value of a column between range .

Syntax :-**Alter table tablename add :-**

Constraint constraints_name check [condition] ;

EX :- Alter table emp add constraint c2 check (sal > 5000) ;

Syntax :-**Drop check using Alter :-**

Alter table tablename drop constraint constraint_name ;

EX :- Alter table emp drop constraint c2 ;

14.Explain about relational set operators in SQL

A. **Set Operators:** SQL supports few set operators to be performed on table data.

These are used to get meaningful results from data under different special conditions.

1. Relational set operators are used to combine or subtract records from two tables.
2. These operators are used in the select query to combine or remove the records.
3. In order to perform set operators the two tables should have same no of columns and datatypes.

The different types of set operators are:

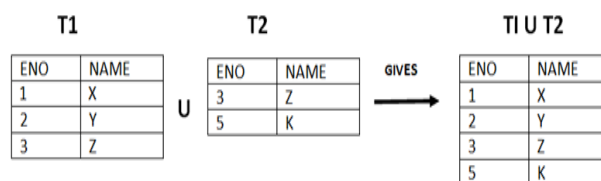
Operators	Returns
Union	All distinct rows selected by either query.
Union all	All rows selected by either query including all duplicates.
Intersect	All common rows are selected by both queries.
Minus	All distinct rows selected by the first query but not the second.

1.UNION:

- It is used to combine the result of two or more select statements.
- It will eliminate duplicate rows from its result set.
- In union, number of columns and datatypes must be same in both the tables.

Union query:

select * from T1 UNION select * from T2;



2.Union All: This operation is similar to union, but it also shows duplicate rows.

Union All query:

select * from T1 UNION ALL select * from T2;

3.Intersect:

- This operation is used to combine two select statements, but it only returns records which are common for both the select statements.
- It picks only duplicate records from tables.
- In intersect, the number of columns and datatypes must be same in both the tables.
- MySQL doesnot support intersect operator.

Intersect Query:

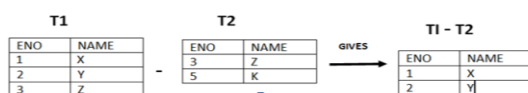
select * from T1 INTERSECT select * from T2;

**4.Minus:**

This operator is used to display the records that are present only in the first table and doesnot present in the second table.

Minus Query:

select * from T1 MINUS select * from T2;

**15. Write about query, subquery and correlated subquery in SQL?****A. Query :**

1. A query is a question.
2. A query is request for a table to retrieve some useful information from the table.
3. Different query languages are used to frame queries.

Basic SQL query :

The basic form of SQL query is as follows:

Syntax :

select [Distinct] [*] column_name(s) from tablename where [condition];

Subquery :

1. A query with in a query is called as subquery.
2. It is also known as Nested Query or Inner Query.
3. A subquery is a query with in another SQL query and embedded with in the where clause.
4. Subqueries are nested queries that provide data to the enclosing query.
5. In subqueries, the main query is called as parent query or outer query.

Rules :

1. Subqueries must be enclosed with paranthesis.
2. It can be used with select, insert, update and delete statements along with the operators like “= ,< , > , >= , in, between” etc.
3. It can return individual values or a list of records.

Subqueries with “select” statement :

- Subqueries are most frequently used with the select statement.
- The basic syntax is as follows:

Syntax :

Select column_name(s) from tablename where value in (select column_names from tablename where condition);

In subquery processing, first the inner query is executed and then the inner query result is acting as input to the output query.

Ex : 1. emp table :

eno	Ename	sal	dno
1	Vijju	30000	10
2	Lakshmi	20000	20
3	Alekhyia	25000	30

Subquery :

SQL > select * from emp where sal > (select sal from emp where eno=2);

Result : eno ename sal dno

2 lakshmi 20000 20

2.Subquery :

SQL > select * from emp where dno = (select dno from emp where eno=3);

Result : eno ename sal dno

3 alekhyia 25000 30

3.dept table :

dno	Dname
10	Computers
20	Maths

Subquery :

SQL > select * from emp where dno = (select dno from dept where dname like 'computers');

Result : eno enamesaldno

1	vijju	30000	10
---	-------	-------	----

Correlated subquery :

A correlated query is a subquery which uses values from the outer query in its where clause.

16. Write about Embedded SQL?**A. SQL:-**

*Structure Query Language(SQL) is standardized language used to manipulate database objects and the data they contain.

*It is comprised of several different statements that are used to manipulate data value.

*SQL is a non-procedural programming language.

Ex:-SQL>select * from emp.

Embedded SQL:-

*It is a method of combining the computing power of programming language and the database manipulation capabilities of SQL.

*As a result, database applications are usually developed by combining capabilities of a high level programming language with SQL.

*The simplest approach is to embed SQL statements directly into the source code files that will be used to create an application.

*This technique is referred to as embedded SQL programming.

*High level programming language compilers cannot interpret the SQL statements.

*Hence, source code files containing embedded SQL statements which must be preprocessed before compiling.

*These statements are written line with in the program source code of the host language.

*Embedded SQL statements are written in any number of languages such as C, C++, Java, COBOL, FORTRAN etc.,

*The popular host language for embedded SQL is C

*SQLCAH-Structure Query language communication area header file to be included in C.

*Each SQL statements coded in a high level programming language source code file must be prefixed with the key words EXEC SQL and END EXEC

Syntax:

```
EXEC SQL
    <embedded SQL statements>
END EXEC
```

*The statements starts with EXEC SQL and terminated with either a semi colon(;) (or)the keywords END EXEC.

Host Variables:-

*The database manager cannot work directly with high level programming language variable.

*So, it must use special variables known as Host variables.

*These are used to move data between an application and a database.

*There are two types of host variables.

Input host variables:-Transfer data to database

Output host variables:-Receives data from database.

*They must be defined with a special section known as Declare Section

*Each host variables must be accessed a unique name

EX:- C program with embedded SQL

```
#include<stdio.h>
```

```
EXEC SQL include SQL CA
```

```
main( )
```

```
{
```

```
EXEC SQL BEGIN DECLARE SECTION
```

```
Int customer id, ordered;
```

```
Char salesperson[10]; status[6];
```

```
EXEC SQL END DECLARE SECTION;
```

```
printf("Enter order number");
```

17. Write about selection and projection?**Select:-**

- The select operator is used to retrieve horizontal subset of all records from a table.
- This operator is used to retrieve all the rows found in a table that satisfies a specified conditions.
- If the condition is not applied for select statement ,then it returns all the rows from the table.
- Here where clause specifies the conditions

EX:-Emp table

Eno	Ename	Sal	dno
1	Ali	3000	10
2	Taraka	4000	20
3	Madhu	2000	30

SQL>Select * from emp where eno<4;

Output:-

Eno	EName	Sal	dno	
1	Ali	3000	10	
2	Taraka	4000	20	
3	Madhu	2000	30	

Project:-

- The project operator gives all values for selected columns.
- This operator gives vertical subset of all records from a table.
- This operator selects the column which we want in a query.

EX:-SQL>select eno,ename from emp;

Output:- Eno Ename

- 1 Ali
- 2 Taraka
- 3 Madhu
- 4 Ex:-SQL>select eno , sal from emp;

